AI-POWERED PASSPORT APPLICATION ASSISTANCE AND OPAS INTELLIGENCE

Expression of Interest (EOI)

UNIPODS AI for Public Good Challenge

1. Executive Summary

The Ministry of Home Affairs and Internal Security (MoHAIS) proposes an AI-powered enhancement to the existing Online Passport Application System (OPAS) to improve the passport application experience for citizens and increase operational efficiency within the Passport Office.

The proposed solution will introduce an intelligent assistance layer that guides applicants through the application process, validates contact information, identifies incomplete applications, detects paid-but-unsubmitted applications, and automates reminders for applicants who need to take further action.

The system will also provide operational intelligence dashboards to help passport officers and administrators monitor application progress, identify bottlenecks, track reverted cases, and make informed, data-driven decisions.

The proposed initiative will be implemented as a six-month pilot, using approved OPAS data and existing workflows. Rather than replacing or redeveloping OPAS, the project will enhance its capabilities through AI-assisted guidance, data validation, automated notifications, and operational analytics.

2. Background and Problem Statement

The Online Passport Application System provides a digital channel through which citizens can apply for passports. However, several challenges affect application completion, communication with applicants, and the efficiency of passport processing.

The key challenges include:

- Incomplete applications: Applicants may leave required fields incomplete or submit incorrect information.
- Invalid contact information: Incorrect email addresses and phone numbers can prevent applicants from receiving important updates.
- Paid but unsubmitted applications: Some applicants make payments but do not complete or submit their applications.
- Reverted applications: Approximately 1,500 cases require further action from applicants, creating a significant follow-up burden.
- High enquiry volumes: Passport officers spend considerable time responding to enquiries and following up on outstanding applications.
- Limited operational visibility: Officers and administrators need effective ways to identify outstanding cases, monitor application progress, and understand recurring service bottlenecks.

These challenges can contribute to delays, repeated communication, increased administrative workload, and an unsatisfactory experience for citizens. There is therefore an opportunity to enhance OPAS with AI-assisted tools that help applicants complete their applications correctly while providing officers with better information for managing the application process.

3. Proposed AI Solution

The project proposes an AI-assisted OPAS enhancement layer that works alongside the existing system. It will combine an applicant-facing virtual assistant, data validation mechanisms, automated notifications, and an operational intelligence dashboard.

3.1 Applicant Guidance Assistant

A virtual assistant will help applicants understand the passport application process by providing guidance on requirements, procedures, common errors, and outstanding actions. Its capabilities will include answering frequently asked questions, explaining requirements in simple language, guiding applicants through relevant stages, helping applicants understand why an application may have been reverted, and directing applicants to an appropriate official channel when human assistance is required.

3.2 Contact Information and Application Data Validation

The solution will identify potentially invalid or incorrectly formatted contact details and application information. It will support email-format validation, phone-number format and country-code validation, detection of missing required fields, identification of incomplete or inconsistent application information, and clear prompts explaining what applicants need to correct. Where verification services are available and approved, additional checks may be considered. Format validation alone will not be treated as proof that a phone number or email address belongs to the applicant.

3.3 Incomplete and Unsubmitted Application Detection

The system will analyse available OPAS workflow data to identify applications that require further action, including incomplete draft applications, paid applications that have not been submitted, applications awaiting corrections or additional information, and applications that have remained inactive beyond defined periods. Applicants will receive appropriate prompts or reminders, subject to approved notification rules and available contact information.

3.4 Automated Reminders and Notifications

The solution will support timely communication with applicants whose applications require action. Notifications may include reminders to complete and submit applications, alerts about reverted applications, guidance on outstanding corrections, and notifications through approved email, SMS, or OPAS message-board channels. Delivery and applicant responses will be tracked where supported. The notification process will follow approved communication schedules and privacy requirements to avoid unnecessary or duplicate messages.

3.5 Operational Intelligence Dashboard

An intelligence dashboard will give authorised Passport Office personnel and administrators a consolidated view of application progress and outstanding actions. It will support monitoring of total applications by workflow status, incomplete and paid-but-unsubmitted applications, reverted cases, invalid or unverified contact details, reminder delivery and response rates, application completion trends, and recurring issues or potential processing bottlenecks. These insights will help management prioritise follow-ups, allocate resources, and identify opportunities to improve service delivery.

4. Objectives

4.1 General Objective

To develop and pilot an AI-assisted enhancement to OPAS that improves passport application completion, citizen communication, and operational decision-making.

4.2 Specific Objectives

- Provide applicants with accessible, AI-assisted guidance throughout the passport application process.
- Reduce preventable application errors through automated validation and clear correction prompts.
- Identify incomplete and paid-but-unsubmitted applications that require further action.
- Improve the follow-up of reverted applications through automated reminders and tracking.
- Reduce repetitive enquiries and manual follow-up work for passport officers.
- Provide operational dashboards and analytics to support evidence-based management decisions.
- Evaluate the effectiveness, usability, security, and potential scalability of the proposed solution.

5. Pilot Scope

5.1 In Scope

- AI-powered virtual assistant and applicant support.
- Email and phone number format validation.
- Detection of incomplete and paid-but-unsubmitted applications.
- Automated reminders for reverted cases and other outstanding actions.
- OPAS message-board notifications.
- Operational intelligence dashboard.
- Prototype testing, performance evaluation, and pilot reporting.

5.2 Out of Scope

- Complete redevelopment or replacement of OPAS.
- Implementation of a national AI platform.
- Legislative or policy reforms.
- Procurement and deployment of a full production system.
- Integration with unrelated immigration services.
- Automated approval or rejection of passport applications.

6. Data Sources and Management

The pilot will use approved data sources needed to develop, test, and evaluate the proposed solution.

Data source	Intended use
OPAS workflow data	Identify application status, incomplete cases, paid-but-unsubmitted applications, and reverted cases.
Contact information records	Assess email and phone number formats and identify potential communication issues.
Frequently asked questions (FAQs)	Support the virtual assistant's responses.
Enquiry records	Identify recurring questions and common applicant challenges.
Service performance reports	Establish baseline performance and measure improvements.

6.1 Data Protection and Governance

Data access and processing will follow the Ministry's approved governance procedures. The pilot will prioritise anonymised, synthetic, or sandboxed data wherever possible. Safeguards will include role-based access controls, data minimisation and purpose limitation, audit trails for access and system activities, secure storage and controlled data transfer, appropriate retention and deletion procedures, human oversight of sensitive operational decisions, and testing to reduce the risk of exposing personal information through AI-generated responses.

Access to live applicant data, if required, will depend on explicit approval from the relevant data owners and compliance with applicable data protection requirements.

7. Implementation Approach and Timeline

The proposed pilot will run for six months, using an incremental development and evaluation approach.

Week 1 — Requirements and Baseline Assessment: Confirm user needs, review OPAS workflows, agree on success indicators, assess data availability, and establish governance requirements.

Week 2 — Data Preparation and System Design: Prepare approved datasets, define validation rules, design the virtual assistant, and establish the dashboard's data model.

Week 3–4 — Prototype Development: Develop the applicant guidance assistant, validation mechanisms, application-status detection, notification functionality, and operational dashboard.

Week 5 — Testing and Evaluation: Test functionality, data protection, response quality, notification rules, dashboard accuracy, and usability with relevant stakeholders.

Week 6 — Pilot Demonstration and Reporting: Demonstrate the prototype, compare performance against the baseline, document lessons learned, and recommend next steps.

Note: The timeline is proposed and will be refined with MoHAIS and the UNIPODS programme team.

8. Expected Deliverables

1. **Applicant Guidance Assistant:** A working prototype that answers approved FAQs and guides applicants through common application issues.
2. **Data Validation Module:** Mechanisms for identifying invalid contact formats, missing fields, and potential application errors.
3. **Application Monitoring Module:** Logic to identify incomplete, paid-but-unsubmitted, and reverted applications requiring attention.
4. **Automated Notification Prototype:** Functionality for generating and tracking approved reminders and message-board notifications.
5. **Operational Intelligence Dashboard:** A dashboard showing application status, outstanding cases, completion trends, and key performance indicators.
6. **Testing and Evaluation Report:** Findings on prototype accuracy, usability, operational value, and limitations.
7. **Implementation Roadmap:** Recommendations for improving, integrating, and potentially scaling the solution after the pilot.

9. Success Indicators and Evaluation Framework

The pilot will be assessed against measurable indicators, using an agreed baseline and targets established with the Ministry during the first month.

Evaluation area	Indicator	Measurement approach
Contact validation	Invalid-format contact details identified	Compare results against a verified test dataset.
Application completion	Incomplete and paid-but-unsubmitted cases detected	Compare detected cases against known workflow records.
Reverted case management	Outstanding reverted cases identified and reminders generated	Track eligible cases, reminders, and subsequent applicant actions.
Citizen assistance	Accuracy and usefulness of guidance	Evaluate responses against approved FAQs and test scenarios.
Operational efficiency	Changes in manual follow-up and repetitive enquiries	Compare baseline and pilot-period workload measures.
Dashboard performance	Accuracy of application status and operational summaries	Reconcile dashboard outputs against approved source records.
Citizen experience	Applicant satisfaction and ease of use	Use structured feedback or usability testing.

Targets will be agreed upon before testing begins. Where the prototype operates in a sandbox, results will be reported as technical validation findings rather than claimed improvements in live service delivery.

10. Governance, Ownership, and Stakeholder Responsibilities

The project will operate under the Ministry's established governance and oversight arrangements.

Role	Responsible party
Executive Sponsor	Permanent Secretary, MoHAIS
Programme Focal Point	Spriano Banda, Chief Systems Development Officer
Business and Operational Stakeholders	Passport Office
Technical and Data Stakeholders	ICT Department, OPAS technical administrators, and data administrators
Pilot Development and Evaluation	Designated project team, working with relevant Ministry personnel and programme mentors

10.1 Governance Principles

- The Ministry retains oversight of the pilot and its operational requirements.
- Data access is approved by the relevant data owners.
- AI outputs support applicants and officers but do not replace authorised decision-makers.
- Sensitive cases are referred to appropriate human personnel.
- Testing, performance, and security findings are documented for review.

11. Expected Benefits and Strategic Value

11.1 Benefits to Citizens

- Clearer guidance on passport application requirements.
- Fewer preventable errors and incomplete submissions.
- Better awareness of outstanding actions.
- More timely communication about reverted applications.
- Improved access to information about the application process.

11.2 Benefits to the Passport Office

- Reduced repetitive enquiries and manual follow-ups.
- Earlier identification of applications requiring attention.
- Improved visibility of outstanding and reverted cases.
- Better prioritisation of staff time and operational resources.
- Data-driven insights into recurring application challenges.

11.3 Strategic Value to Government

The project supports citizen-centred public service delivery, digital transformation, responsible AI adoption, and evidence-based decision-making. If the pilot demonstrates sufficient operational value, the approach could inform future enhancements to other government digital services, including immigration services, national registration, citizenship services, and other online application platforms. Any extension would require separate approval, requirements assessment, and data governance review.

12. Risks and Mitigation Measures

Potential risk	Proposed mitigation
Incomplete or poor-quality data	Assess data quality early and document limitations.
Unauthorised access to personal information	Apply approved access controls, data minimisation, and audit trails.
Incorrect AI-generated guidance	Ground responses in approved information and provide escalation to human officers.
Incorrect reminders or duplicate notifications	Define clear eligibility rules, test notification logic, and maintain delivery logs.
Limited access to OPAS integration interfaces	Begin with approved sandbox data and use simulated workflows where necessary.
Overreliance on AI outputs	Keep final operational and passport decisions under authorised human control.
Low adoption by applicants or officers	Involve users in design, usability testing, and pilot evaluation.

13. Sustainability and Potential for Scale

The proposed solution is designed to complement existing OPAS workflows rather than introduce a replacement system. Its modular design will allow individual capabilities—such as applicant guidance, data validation, reminders, and analytics—to be assessed and improved independently.

Following the pilot, MoHAIS can use the evaluation findings to determine whether to proceed with further integration, strengthen the prototype, or extend selected capabilities to other government services. Any wider deployment will depend on demonstrated value, technical feasibility, funding, security review, and formal approval.

14. Conclusion

The proposed AI-Powered Passport Application Assistance and OPAS Intelligence initiative offers a practical opportunity to improve passport application completion, communication, and operational efficiency within the Ministry of Home Affairs and Internal Security.

By combining intelligent applicant guidance, contact and application data validation, automated reminders, and operational dashboards, the project seeks to address specific challenges within the existing passport application process.

The six-month pilot is intended to produce a functional, measurable, and security-conscious prototype that uses existing workflows and approved data rather than requiring a complete redevelopment of OPAS.

The central proposition is simple: help citizens complete their passport applications correctly, help officers identify cases requiring attention, and help management make better decisions using reliable operational information.